

Alpilla 600 MW Combined Cycle Power Plant

Engineering execution plan (workflow, interfaces, timeline)

Document number	ALP-EPC-00000-GE-PLN-0001
Revision	draft - not yet issued (next issue: IFR rev A)
Originator	Istanbul EPC
Review class	approval
Planned issue (workflow timeline)	IFR 2027-01-18, IFA 2027-02-15, IFC 2027-02-22
Basis	requirement:ER-01.09

1. Purpose and scope

This plan sets how the engineering of the project is executed: scope (the MDL), organisation, the workflow between documents and disciplines, the review cycle and the timeline against the construction need dates (Advanced Work Packaging, ALP-EPC-00000-GE-PLN-0002). All figures are taken from the MDL and the workflow timeline computed from it; they change when the records change.

2. Organisation and responsibilities

Party	Responsibility
Istanbul EPC	Consortium leader: plant design and balance of plant engineering, integration of the power island, MDL and document control, procurement of BOP packages, construction and commissioning management
Imaginary Electric	Power island design and supply (GT, HRSG, ST, generators, GSUs, TCS); VDRL documents per the MDL dates (TQ-IEC-023); interface data (IF- points)
Owner	Review and approval per the review class (14 / 21 days); approval of the KKS key list, numbering and AWP (TQ-035)
Engineer	Owner's Engineer: technical review on behalf of the Owner, KKS key verification

3. Engineering scope (MDL)

Discipline	Docs	EPC	Supplier	BDP	In EWP	Hours	First IFR	Last IFC
civil	42	39	3	1	28	3,072	2026-11-02	2027-08-27
commissioning	11	11	0	0	0	960	2026-11-23	2027-09-27
electrical	82	76	6	2	23	5,215	2026-09-28	2027-06-14
general	15	12	3	2	0	1,440	2026-09-28	2027-03-22
hse	16	15	1	0	0	2,420	2026-11-23	2027-10-07
hvac	11	11	0	0	3	871	2026-11-02	2027-03-12
i and c	88	84	4	2	17	6,020	2026-11-02	2027-10-20
mechanical	155	143	12	0	5	6,620	2026-09-28	2028-07-24
piping	89	87	2	3	61	11,556	2026-11-02	2027-10-22
process	98	92	6	12	0	9,825	2026-09-28	2027-08-09
project control	2	2	0	0	0	160	2027-01-11	2027-03-29
structural	14	14	0	0	9	1,158	2026-11-02	2027-06-04

Total 623 documents, 49,317 engineering hours (document preparation; 3D modelling, vendor document review and site support are budgeted separately).

Coverage is checked at every engine run: every KKS system has the document types its category requires, every Employer's Requirement and every EPC / IEC scope item is answered by at least one document, every equipment item has a datasheet, an MR and a CWP, every MR a specification and a requisition, every CWP an EWP.

4. Workflow rules

- Documents form a network through their inputs (MDL column Inputs). A document starts when all its inputs have reached the maturity its type requires (IFR or IFC, section 4 of ALP-EPC-00000-GE-PRC-0002), not before NTP.
- First issue (IFR) = start + preparation time of the type (+5 % per extra sheet, at most double).
- Owner review 10 wd (15 wd for the Basic Design Package); approval class: IFA = IFR + review + update, IFC = IFA + 5 wd; review class: IFC = IFR + review + update; information / internal: IFC = IFR + update.
- Tender-stage documents (bid design) are issued at NTP. Supplier documents keep their committed VDRL dates. Issued revisions replace the planned dates.

- Construction documents belong to the EWP of their CWP; the EWP is ready when its last document is IFC and is needed ewp_lead_days before the CWP starts. Requisitions start the PWP (MR) cycle: PO = MRQ IFC + bid + award.

5. Interdisciplinary inputs (who feeds whom)

Number of input links from the supplying discipline (row) to the receiving discipline (column).

from \ to	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS
CV	55		1									16
CM		3										
EL	1	12	128									
GE	1		1	11	2	2	1	3	3	42	3	
HS		1			10		1					1
HV	6	3				13	1		1	1		
IC		5					105	1				
ME	1	36			1		1	164	4			1
PI	27		13		7	4	8	1	214	41		3
PR	3	103	1		40		81	61	86	130		
PC												
SS	1							1				7

Typical inputs per document type (checked at every run; a document without any of them is reported):

Type	Title	Inputs at	Typical inputs
ARC	Architectural drawing	IFR	LAY
BLK	Block / architecture diagram	IFR	DBR
CAL	Calculation	IFR	SDS, SLD, STU, PID, DBR, CAL, HMB
CBL	Cable schedule and routing	IFR	SLD, LAY
CMP	Commissioning procedure	IFR	PID, CNR, CAL, STU, SPC
CNR	Control narrative / functional description	IFR	PID, CNR
CVL	Civil drawing (grading, roads, drainage, marine)	IFR	PLT, STU, SDS
DSH	Datasheet	IFR	SPC
ELY	Electrical layout (earthing, lighting, trays, equipment)	IFR	LAY, SLD
FND	Foundation / underground drawing	IFR	LAY, CAL
GAD	General arrangement drawing	IFR	LAY
HAZ	Safety study (HAZID, HAZOP, SIL, hazardous areas, fire, QRA)	IFR	PID, PLT, HAZ
HMB	Heat and mass balance	IFR	DBR
IOL	I/O list	IFR	PID, CNR, LST, IOL
ISO	Isometric drawing package	IFR	PPL
ITP	Inspection and test plan	IFR	SPC
LAY	Equipment layout / arrangement	IFR	PLT
LOG	Logic diagram	IFR	CNR, IOL
LOP	Loop / connection diagram	IFR	IOL
LST	List / index (equipment, lines, valves, instruments, loads, tie-ins)	IFR	PID, PFD, LST, SLD, DBR, PRC
M3D	3D model review report (30 / 60 / 90 %)	IFR	PLT, LAY, PPL, ISO
MRQ	Material requisition	IFR	SPC
OMM	Operation and maintenance manual	IFR	PID
PFD	Process flow diagram	IFR	HMB
PID	Piping and instrumentation diagram	IFR	SDS
PLN	Plan (execution, quality, HSE, commissioning, training)	IFR	PRC, PLN, DBR, PFD, LST
PLT	Plot plan	IFR	DBR
PPL	Piping general arrangement / plans	IFR	PID, LAY
SCH	Schedule	IFR	MDL
SDS	System design description	IFR	DBR
SLD	Single line diagram	IFR	LST
STL	Steel structure drawing	IFR	GAD, CAL
STR	Pipe stress analysis report	IFR	PPL

6. Timeline

IFC / final issues per quarter and discipline (the monthly curve and the AWP bars are in ALP_engineering_timeline.pdf of the engineering_plan engine):

Quarter	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS	Total
2026 Q4	11	1	29	7	1	6	16	97	5	46		5	224
2027 Q1	27	2	32	8	8	5	27	55	32	19	2	4	221
2027 Q2	3	4	21		2		22		26	24		5	107
2027 Q3	1	4			4		20	1	25	9			64
2027 Q4					1		3	1	1				6
2028 Q3								1					1

Document preparation capacity (eng_resource, levelled timeline): CV 5 FTE, CM 1 FTE, EL 4 FTE, GE 2 FTE, HS 2 FTE, HV 1 FTE, IC 5 FTE, ME 8 FTE, PI 8 FTE, PR 4 FTE, PC 1 FTE, SS 2 FTE - total 43 FTE. Documents are scheduled in the order of their latest start (backwards from the EWP and PWP need dates) within these capacities.

Engineering load per month: hours of each EPC document as booked by the levelled timeline (spread over its preparation when not levelled); 143 tender-stage documents (8,589 h) were prepared in the bid and are excluded; FTE at 150 h / month.

Month	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS	Hours	FTE
2026-11	278	149	630	315	315	157	649	1,260	1,260	630		48	5,691	38
2026-12	727	135	690	322	342	172	539	1,380	966	690	48		6,012	40
2027-01	569	36	630	315	238	146	356	963	753	630	32	150	4,818	32
2027-02	564	72	600	68	145	39	360	1	646	600	53	300	3,448	23
2027-03	84	108	690		8		392		1,287	690	27	333	3,618	24
2027-04		39	573		197		352		1,320	660		87	3,228	22
2027-05		27	35		259		396		1,260	630			2,607	17
2027-06	73	129			316		271		1,320	550			2,658	18
2027-07	77	165			330		664		1,312	60			2,608	17
2027-08		100			204		825		887				2,016	13
2027-09					6		430		105				541	4
2027-10							95						95	1

Peak 2026-12: 6,012 h (about 40 FTE).

Key dates (CPM, driven by mechanical completion of the CWP):

Key date	Title	Forecast	Float (wd)
KD-000	Notice to Proceed	2026-11-02	457
KD-010	Back-feed energisation (key date)	2028-07-04	21
KD-020	GT first fire (key date)	2028-09-18	10
KD-030	First synchronisation in combined cycle (key date)	2028-11-20	31
KD-040	Taking-Over of the Works (Time for Completion, 975 days)	2029-04-02	67

7. Power island supplier documents (VDRL)

Number	IEC ref	Title	IFR	IFC	Feeds
ALP-IEC-00000-GE-LST-0001	IEC-ALP-IF-001	Terminal points and interface data - power island	2026-09-28	2026-11-02	1
ALP-IEC-00000-GE-RPT-0001	IEC-ALP-DOR-001	Division of responsibility - power island and balance of plant	2026-09-28	2026-11-02	0
ALP-IEC-00000-GE-SPC-0001	IEC-ALP-REQ-001	Power island interface requirements to the EPC	2026-09-28	2026-11-02	0
ALP-IEC-10BAT-EL-DSH-0001	IEC-ALP-DS-106	Generator step-up transformers datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10BAT-EL-GAD-0001	IEC-ALP-GA-104	Generator step-up transformers - GA, installation and maintenance	2026-09-28	2027-01-11	1
ALP-IEC-10HAA-ME-DSH-0001	IEC-ALP-DS-104	HRSG IE-HR3 datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10HAA-ME-GAD-0001	IEC-ALP-GA-103	HRSG IE-HR3 - GA, erection and maintenance	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-ME-DSH-0001	IEC-ALP-DS-101	Gas turbine IE-9H.02 datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10MBA-ME-GAD-0001	IEC-ALP-GA-101	Gas turbine and generator - GA, maintenance and installation	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-ME-LAY-0001	IEC-ALP-GA-100	Power island arrangement - installation and maintenance spaces	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-PR-HMB-0001	IEC-ALP-PER-001	Power island thermal performance data (heat balance)	2026-09-28	2026-11-02	1
ALP-IEC-10MBA-PR-HMB-0002	IEC-ALP-PER-003	Power island off-design heat balances (case book)	2026-09-28	2026-11-02	0
ALP-IEC-10MKA-EL-DSH-0001	IEC-ALP-DS-102	Generators, GCB and isolated phase busduct datasheet	2026-09-28	2026-12-28	1
ALP-IEC-20MAA-ME-DSH-0001	IEC-ALP-DS-103	Steam turbine IE-ST3R datasheet	2026-09-28	2026-12-28	0
ALP-IEC-20MAA-ME-GAD-0001	IEC-ALP-GA-102	Steam turbine, generator and condenser - GA, maintenance	2026-09-28	2027-01-11	1
ALP-IEC-20MAG-ME-DSH-0001	IEC-ALP-DS-105	Seawater-cooled surface condenser datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10HAA-CV-CAL-0001	IEC-ALP-FL-102	Foundation load data and anchor bolt drawings - HRSG	2026-12-14	2027-01-11	2
ALP-IEC-10MBA-CV-CAL-0001	IEC-ALP-FL-101	Foundation load data and anchor bolt drawings - GT and generator	2026-12-14	2027-01-11	3
ALP-IEC-00000-PI-LST-0001	IEC-ALP-IF-001-A	Terminal point nozzle loads (IF-001 attachment A)	2026-12-28	2027-01-25	9
ALP-IEC-10BAA-EL-SLD-0001	IEC-ALP-SLD-101	Power island single line diagram (generators, GCB, GSU, UAT tap-offs, SFC)	2026-12-28	2027-02-08	1
ALP-IEC-10MBA-EL-LST-0001	IEC-ALP-EL-101	Power island electrical load list	2026-12-28	2027-02-08	1
ALP-IEC-20MAA-CV-CAL-0001	IEC-ALP-FL-103	Foundation load data - ST, generator and condenser	2026-12-28	2027-01-25	3
ALP-IEC-10HAA-PR-PID-0001	IEC-ALP-PID-102	HRSG P&IDs incl. SCR and ammonia injection	2027-01-11	2027-03-08	2
ALP-IEC-10MBA-HS-RPT-0001	IEC-ALP-HS-101	Power island noise data (ISO 10494)	2027-01-11	2027-02-22	1
ALP-IEC-10MBA-PR-PID-0001	IEC-ALP-PID-101	Gas turbine P&IDs (fuel gas, liquid fuel, lube oil, inlet air, washing)	2027-01-11	2027-03-08	2
ALP-IEC-10MBY-IC-BLK-0001	IEC-ALP-IC-101	TCS architecture and DCS integration (OPC UA, IEC 62443 zone)	2027-01-11	2027-02-22	1

Number	IEC ref	Title	IFR	IFC	Feeds
ALP-IEC-10MBA-ME-ITP-0001	IEC-ALP-QA-101	Power island inspection and test plans	2027-01-25	2027-02-22	0
ALP-IEC-10MKA-EL-RPT-0001	IEC-ALP-EL-102	Generator capability curves and dynamic models (Grid Code)	2027-01-25	2027-03-22	1
ALP-IEC-10MKA-PR-PID-0001	IEC-ALP-PID-104	Generator auxiliary P&IDs (H2/CO2, seal oil, stator water)	2027-01-25	2027-03-22	0
ALP-IEC-20MAA-PR-PID-0001	IEC-ALP-PID-103	Steam turbine and condenser P&IDs (lube, gland, vacuum, EHC, bypass valves)	2027-01-25	2027-03-22	1
ALP-IEC-10MBY-IC-CNR-0001	IEC-ALP-IC-103	Power island operating and control description (start-up, load, trips)	2027-02-08	2027-04-05	2
ALP-IEC-10MBA-PI-M3D-0001	IEC-ALP-M3D-101	Power island 3D model export (for the plant model)	2027-02-22	2027-04-19	0
ALP-IEC-10MBY-IC-IOL-0001	IEC-ALP-IC-102	TCS I/O list and interface signal list	2027-02-22	2027-04-19	3
ALP-IEC-10MBY-IC-CBL-0001	IEC-ALP-IC-104	Power island cable list (IEC scope and interfaces)	2027-05-31	2027-07-26	2
ALP-IEC-10MBA-ME-PRC-0001	IEC-ALP-ER-101	Power island erection manuals and procedures	2027-07-12	2027-09-06	0
ALP-IEC-10MBA-ME-RPT-0001	IEC-ALP-OM-102	Maintenance concept and long-term service offer	2027-08-09	2027-10-04	0
ALP-IEC-10MBA-ME-OMM-0001	IEC-ALP-OM-101	Power island operation and maintenance manuals	2028-03-06	2028-07-24	0

8. Near-critical items

EWPs with less than 20 working days float (last IFC document drives the EWP):

EWP	CWP	Driving document	Ready	Needed	Float
EWP-05-CV-01	CWP-05-CV-01	ALP-EPC-00UPC-CV-FND-0001	2027-03-24	2027-03-25	1
EWP-03-CV-01	CWP-03-CV-01	ALP-EPC-00UBA-CV-FND-0001	2027-03-24	2027-03-29	3
EWP-11-CV-01	CWP-11-CV-01	ALP-EPC-00UCA-CV-FND-0001	2027-03-24	2027-03-29	3
EWP-06-CV-01	CWP-06-CV-01	ALP-EPC-00UPA-CV-CVL-0001	2027-08-27	2027-09-03	5
EWP-04-CV-01	CWP-04-CV-01	ALP-EPC-00UAC-CV-FND-0001	2027-03-24	2027-04-05	8
EWP-07-CV-01	CWP-07-CV-01	ALP-EPC-00UGA-CV-FND-0001	2027-03-24	2027-04-05	8
EWP-12-CV-01	CWP-12-CV-01	ALP-EPC-00UZA-PI-LAY-0001	2027-01-05	2027-01-18	9
EWP-02-CV-01	CWP-02-CV-01	ALP-EPC-20UMB-CV-FND-0001	2027-03-04	2027-03-22	12
EWP-01-CV-02	CWP-01-CV-02	ALP-EPC-10UHN-CV-FND-0001	2027-03-31	2027-04-19	13
EWP-02-CV-02	CWP-02-CV-02	ALP-EPC-20UMB-CV-FND-0002	2027-03-29	2027-04-19	15
EWP-01-CV-01	CWP-01-CV-01	ALP-EPC-10UMA-CV-FND-0001	2027-02-25	2027-03-22	17
EWP-08-CV-01	CWP-08-CV-01	ALP-EPC-00UQA-CV-FND-0001	2027-04-08	2027-05-04	18

PWPs (MRs) with less than 20 working days float between delivery on site and the first CWP start:

MR	Title	PO	On site	Needed	CWP	Float
MR-ME-101	Gas turbine package incl. auxiliaries (IEC)	2026-11-02	2027-11-15	2027-11-15	CWP-01-ME-01	0
MR-ME-102	HRSG incl. SCR (IEC)	2026-11-02	2027-11-15	2027-11-15	CWP-01-ME-01	0
MR-ME-001	Boiler feed pumps	2027-01-21	2027-11-25	2027-11-29	CWP-08-ME-01	2
MR-ME-003	CW pumps	2027-01-28	2028-01-13	2028-01-17	CWP-05-ME-01	2
MR-ME-009	Auxiliary boiler	2027-01-21	2027-11-25	2027-11-29	CWP-08-ME-01	2
MR-ME-012	Water treatment package (SWRO, demin, condensate polishing)	2027-02-04	2028-01-20	2028-01-24	CWP-07-ME-01	2
MR-ME-015	Gas receiving and metering station	2027-01-21	2027-11-25	2027-11-29	CWP-09-ME-01	2
MR-ME-016	LDO unloading and forwarding / treatment skids	2027-01-21	2027-10-28	2027-11-01	CWP-10-ME-01	2
MR-EL-001	380 kV GIS	2026-12-15	2028-01-25	2028-02-01	CWP-04-EL-01	5
MR-EL-101	Generators, excitation, GCB and busducts (IEC)	2026-11-02	2027-11-08	2027-11-15	CWP-01-ME-01	5
MR-ME-103	Steam turbine and condenser (IEC)	2026-11-02	2028-01-10	2028-01-17	CWP-02-ME-01	5
MR-CV-002	Civil and building works	2026-12-31	2027-01-14	2027-01-25	CWP-12-CV-01	7
MR-EL-009	Cable trays, earthing and lighting materials	2027-01-28	2027-06-03	2027-06-14	CWP-12-EL-01	7
MR-ME-021	EOT cranes	2027-01-28	2027-11-04	2027-11-15	CWP-01-ME-01	7
MR-PI-002	Alloy piping P91 / P22 (main steam, reheat)	2027-02-11	2027-12-30	2028-01-10	CWP-01-PI-01	7
MR-PI-004	GRP CW pipes and fittings	2027-02-11	2027-10-07	2027-10-18	CWP-05-PI-01	7
MR-IC-001	Plant DCS	2027-01-28	2027-12-02	2027-12-20	CWP-11-IC-01	12
MR-EL-003	MV switchgear 10.5 kV	2027-01-21	2027-10-28	2027-11-22	CWP-03-EL-01	17
MR-EL-006	Emergency diesel generator	2027-01-21	2027-10-28	2027-11-22	CWP-03-EL-01	17
MR-PI-001	Piping bulk materials (carbon steel, stainless)	2027-01-28	2027-07-15	2027-08-09	CWP-12-PI-01	17
MR-HV-001	HVAC plant and equipment	2027-02-16	2027-10-12	2027-11-08	CWP-03-HV-01	19

9. Limitations of this revision

- Dates are levelled against the document preparation capacity of section 6 (basis decision:DEC-EPCE-0007); a change of staffing changes the dates at the next run.
- Hours cover document preparation only (see section 3); isometric counts are estimated per piping system.
- KKS key titles are not yet verified against the licensed VGB key catalogue (Owner's Engineer, TQ-035).

Check results (engine)

6 OK, 0 INFO, 0 WARN (checks: coverage MR, coverage equipment, coverage requirement, coverage scope, coverage system, inputs, key date). WARN must be resolved before the next issue; INFO is noted.

Item	Check	Result	Detail
-	coverage requirement	OK	150 of 150 requirements answered
-	coverage scope	OK	88 of 88 EPC/IEC scope items covered